

Problem Set 2 - Macroeconomics

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February 2025

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Literature review

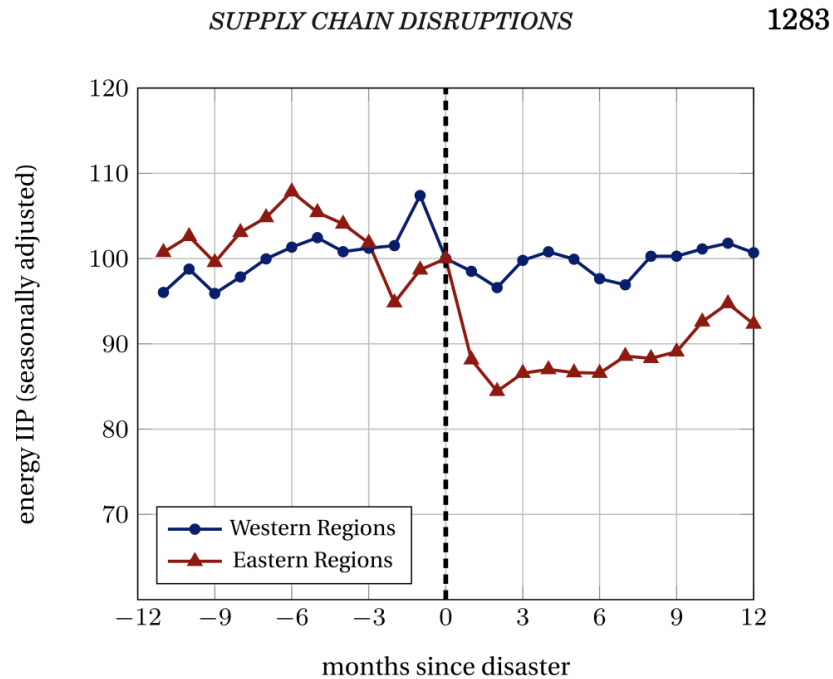


FIGURE IX
Energy Production

Figure 1: Impact of the 2011 Tohoku Earthquake on Japanese energy supply (breakdown East/West)

Figure IX shows the evolution of electricity production in eastern and western Japan around the time of the 2011 earthquake. The line for the east drops sharply after March 2011, reflecting the sudden loss of generating capacity, while the line for the west remains almost perfectly flat. This contrast is important because it rules out the idea that the whole country was hit by a single, uniform energy shortage. Instead, the shock was geographically concentrated, which makes it a genuinely idiosyncratic event. The authors use this fact to strengthen their broader argument. If the fall in output that they observe across many firms were simply due to a nationwide drop in electricity, then their results would not really tell us much about supply-chain transmission. But when they restrict the analysis to firms located in western Japan, which did not suffer power cuts, they still find the same patterns of downstream and upstream propagation. This means that the disruption spreads through firm-to-firm relationships rather than just through one big common shock. The figure therefore plays a crucial role in the paper: it visually separates localized infrastructure damage from the network effects the authors want to measure, and it illustrates how a shock that is small in geographical scope can ripple outward through production linkages.

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Exercise I

2.1 Part A

2.1.1 1)

Lets denote $U = u(c_0) + E(u(c_1)) = u(c_0) + \frac{1}{2}u(c_1^G) + \frac{1}{2}u(c_1^B)$, with $u : c \rightarrow \ln(c)$. The Coefficient of relative risk aversion (CRRA) is given by :

$$CRRA(u) = -c \frac{u''(c)}{u'(c)}$$

Hence the following :

$$CRRA(u) = -c \frac{\frac{-1}{c^2}}{\frac{1}{c}} = 1$$

2.1.2 2)

Let's compute the aggregate income in the bad state of the economy : we denote ω the states of the world and B the bad state

$$\begin{aligned} y_1^B &= E[y|\omega = B] = (1 - \lambda)\mu + \lambda(1 - \frac{\phi}{\lambda})\mu = (1 - \lambda)\mu + \lambda\mu - \phi\mu \\ &= (1 - \phi)\mu \end{aligned}$$

As expected, the aggregate income in this state of the world does not depend on the idiosyncratic risk induced by λ .

2.1.3 3)

We need to consider two separate cases for the budget constraints : the complete market case and the incomplete market case

In the complete market case, the agent is only subject to aggregate risk so he faces the following constraints :

$$\begin{cases} c_0 = \mu - p_2 a_2 - p_1 a_1 - qb \\ c_1^G = \mu + b - a_2 + (1 + \pi_1)a_1 \\ c_1^B = (1 - \phi)\mu + b - a_1 + (1 + \pi_2)a_2 \end{cases}$$

Now let's place ourselves in the incomplete market case : the agent now faces both aggregate and idiosyncratic risk as he cannot fully insure against the latter. Let's denote $c_1^{B,u}$ consumption in the bad state when the agent is not affected (still gets μ) and $c_1^{B,a}$ when the agent is affected (gets $(1 - \frac{\phi}{\lambda})\mu$).

$$\begin{cases} c_0 = \mu - p_2 a_2 - p_1 a_1 - qb \\ c_1^G = \mu + b - a_2 + (1 + \pi_1)a_1 \\ c_1^{B,u} = \mu + b - a_1 + (1 + \pi_2)a_2 \\ c_1^{B,a} = (1 - \frac{\phi}{\lambda})\mu + b - a_1 + (1 + \pi_2)a_2 \end{cases}$$

2.2 Part B

2.2.1 1)

The optimization program is the following :

$$\begin{aligned} & \max_{b, a_1, a_2} \ln(c_0) + \frac{1}{2} \ln(c_1^G) + \frac{1}{2} \ln(c_1^B) \\ \text{s.t. } & \begin{cases} c_0 = \mu - p_2 a_2 - p_1 a_1 - qb \\ c_1^G = \mu + b - a_2 + (1 + \pi_1) a_1 \\ c_1^B = (1 - \phi) \mu + b - a_1 + (1 + \pi_2) a_2 \end{cases} \end{aligned}$$

Substituting the constraints in the objective function, we can write :

$$\max_{b, a_1, a_2} \ln(\mu - p_2 a_2 - p_1 a_1 - qb) + \frac{1}{2} \ln(\mu + b - a_2 + (1 + \pi_1) a_1) + \frac{1}{2} \ln((1 - \phi) \mu + b - a_1 + (1 + \pi_2) a_2)$$

2.2.2 2)

Let's derive the FOC's for the above optimization program : we denote U the objective function

$$\begin{aligned} \frac{\partial U}{\partial b} = 0 & \Leftrightarrow \frac{-q}{c_0} + \frac{1}{2} \left(\frac{1}{c_1^G} + \frac{1}{c_1^B} \right) = 0 \Leftrightarrow q = \left[\frac{1}{c_1^G} + \frac{1}{c_1^B} \right] \frac{c_0}{2} \\ \frac{\partial U}{\partial a_1} = 0 & \Leftrightarrow \frac{-p_1}{c_0} + \frac{1}{2} \left(\frac{1 + \pi_1}{c_1^G} - \frac{1}{c_1^B} \right) = 0 \Leftrightarrow p_1 = \left[\frac{1 + \pi_1}{c_1^G} - \frac{1}{c_1^B} \right] \frac{c_0}{2} \\ \frac{\partial U}{\partial a_2} = 0 & \Leftrightarrow \frac{-p_2}{c_0} - \frac{1}{2} \left(\frac{1}{c_1^G} - \frac{1 + \pi_2}{c_1^B} \right) = 0 \Leftrightarrow p_2 = \left[\frac{1 + \pi_2}{c_1^B} - \frac{1}{c_1^G} \right] \frac{c_0}{2} \end{aligned}$$

We assume that assets are priced considering exogenous consumption profile and equal to the income. Then, we replace c_0 , c_1^G and c_1^B by their values to solve for asset prices in the complete market :

$$\begin{aligned} q &= \frac{1}{2} \left[1 + \frac{1}{1 - \phi} \right] \\ p_1 &= \frac{1}{2} \left[(1 + \pi_1) - \frac{1}{1 - \phi} \right] \\ p_2 &= \frac{1}{2} \left[\frac{1 + \pi_2}{1 - \phi} - 1 \right] \end{aligned}$$

Let's now write the ratio of p_1 and p_2 on q :

$$\begin{aligned} \frac{p_1}{q} &= \frac{(1 + \pi_1) - \frac{1}{1 - \phi}}{1 + \frac{1}{1 - \phi}} = \frac{\pi_1 - \phi(1 + \pi_1)}{2 - \phi} \\ \frac{p_2}{q} &= \frac{\frac{1 + \pi_2}{1 - \phi} - 1}{1 + \frac{1}{1 - \phi}} = \frac{\pi_2 + \phi}{2 - \phi} \end{aligned}$$

Firstly, we notice that both ratios do not depend on λ . This makes sense since, in the complete market setting, idiosyncratic risk does not come into play. Looking at comparative statics in ϕ , we see that $\frac{p_2}{q}$ is increasing in ϕ . Given that the payoff in the bad state depends negatively on ϕ and that a_2 provides utility in this state, the agent will be willing to pay more to get a greater payoff in the bad state when ϕ goes up. For the ratio involving p_1 the effect of ϕ appears to be ambiguous at first, let us compute its derivative to investigate whether one effect dominates :

$$\frac{\partial \frac{p_1}{q}}{\partial \phi} = \frac{\pi_1 - \phi(1 + \pi_1)}{(2 - \phi)^2} - \frac{1 + \pi_1}{2 - \phi} = \frac{-(2 + \pi_1)}{(2 - \phi)^2} < 0$$

Thus, $\frac{p_1}{q}$ decreases in ϕ . This is in line with the fact that a_1 provides more utility in the good state and is thus more valuable when ϕ goes down (bad state is less punishing).

2.3 Part C

2.3.1 1)

As we must now account for idiosyncratic risk, the optimization program is now the following :

$$\begin{aligned} \max_{b, a_1, a_2} & \ln(c_0) + \frac{1}{2} \ln(c_1^G) + \frac{1}{2} [\lambda \ln(c_1^{B,a}) + (1-\lambda) \ln(c_1^{B,u})] \\ \text{s.t.} & \begin{cases} c_0 = \mu - p_2 a_2 - p_1 a_1 - qb \\ c_1^G = \mu + b - a_2 + (1 + \pi_1) a_1 \\ c_1^{B,u} = \mu + b - a_1 + (1 + \pi_2) a_2 \\ c_1^{B,a} = (1 - \phi) \mu + b - a_1 + (1 + \pi_2) a_2 \end{cases} \end{aligned}$$

As in part B, we substitute the constraints and get :

$$\begin{aligned} \max_{b, a_1, a_2} & \ln(\mu - p_2 a_2 - p_1 a_1 - qb) + \frac{1}{2} \ln(\mu + b - a_2 + (1 + \pi_1) a_1) \\ & + \frac{1}{2} [\lambda \ln((1 - \phi) \mu + b - a_1 + (1 + \pi_2) a_2) + (1 - \lambda) \ln(\mu + b - a_1 + (1 + \pi_2) a_2)] \end{aligned}$$

2.3.2 2)

Let's compute the FOC's as we did in part B :

$$\begin{aligned} \frac{\partial U}{\partial b} = 0 & \Leftrightarrow q = \frac{1}{2} \left[1 + \frac{\lambda^2}{\lambda - \phi} + (1 - \lambda) \right] \\ \frac{\partial U}{\partial a_1} = 0 & \Leftrightarrow p_1 = \frac{1}{2} \left[\pi_1 - \frac{\lambda^2}{\lambda - \phi} + \lambda \right] \\ \frac{\partial U}{\partial a_2} = 0 & \Leftrightarrow p_2 = \frac{1}{2} \left[-1 + (1 + \pi_2) \left\{ \frac{\lambda^2}{\lambda - \phi} + (1 - \lambda) \right\} \right] \end{aligned}$$

This gives us the following ratios:

$$\begin{aligned} \frac{p_1}{q} &= \frac{\pi_1 - \frac{\lambda^2}{\lambda - \phi} + \lambda}{1 + \frac{\lambda^2}{\lambda - \phi} + (1 - \lambda)} = \frac{\lambda(\pi_1 - \phi) + \pi_1 \phi}{\lambda(2 + \phi) - 2\phi} \\ \frac{p_2}{q} &= \frac{-1 + (1 + \pi_2) \left\{ \frac{\lambda^2}{\lambda - \phi} + (1 - \lambda) \right\}}{1 + \frac{\lambda^2}{\lambda - \phi} + (1 - \lambda)} = \frac{\lambda\phi(1 + \pi_2) + \pi_2(\lambda - 1)}{2\lambda + \phi(\lambda - 2)} \end{aligned}$$

As the effect of λ and ϕ is ambiguous for both ratios, we compute the derivatives and get :

$$\begin{aligned} \frac{\partial \frac{p_1}{q}}{\partial \lambda} &= \frac{(\pi_1 - \phi)(\lambda(2 + \phi) - 2\phi) - (2 + \phi)(\lambda(\pi_1 - \phi) + \pi_1 \phi)}{(\lambda(2 + \phi) - 2\phi)^2} = \frac{-2\phi\pi_1 + 2\phi^2 - 2\phi\pi_1 - \pi\phi^2}{(\lambda(2 + \phi) - 2\phi)^2} = \frac{-4\phi\pi_1 + (2 - \pi_1)\phi^2}{(\lambda(2 + \phi) - 2\phi)^2} \\ \frac{\partial \frac{p_1}{q}}{\partial \phi} &= \frac{\lambda(2\pi_1 - \lambda^2)}{(\lambda(2 + \phi) - 2\phi)^2} \\ \frac{\partial \frac{p_2}{q}}{\partial \lambda} &= \frac{\pi_2[\pi_2(1 - \frac{\phi}{2} - \phi^2) - \phi^2]}{(\lambda(2 + \phi) - 2\phi)^2} \\ \frac{\partial \frac{p_2}{q}}{\partial \phi} &= \frac{\lambda^2(2 + \pi_2) + 3\lambda\pi_2 - 2\pi_2}{(\lambda(2 + \phi) - 2\phi)^2} \end{aligned}$$

This result is foreseeable as homographic functions are always monotone on each interval where they are defined. The sign of the derivative depends only on the numerator as the denominator is always positive, and the value of the numerator is independent of the variable of interest. We cannot conclude decisively on how those ratios vary as a function of λ and as a function of ϕ as the effect is ambiguous and depends on other parameters.

3

Exercise II

3.1 1)

We have the following optimization program :

$$\begin{aligned} \max_{a_{t+1}, l_t} \sum_{t=0}^{+\infty} \beta^t u(c_t - hc_{t-1}, l_t) \text{ with } h < 1 \\ \text{s.t } c_t + a_{t+1} = (1 + r_t)a_t + wl_t \end{aligned}$$

Substituting the constraint in the objective function we get :

$$\max_{a_{t+1}, l_t} \sum_{t=0}^{+\infty} \beta^t u((1 + r_t)a_t + wl_t - a_{t+1} - hc_{t-1}, l_t)$$

Now, let's write down the Bellman equation of this problem with the value function :

$$V(a, c_{-1}) = u(\underbrace{(1 + r)a + wl - a' - hc_{-1}}_{=c - hc_{-1}}, l) + \beta V(a', \underbrace{(1 + r)a + wl - a'}_{=c})$$

3.2 2)

To solve for this let's get the Euler equation :

- We get the FOC's with respect to the control variables a', l :

$$\begin{aligned} (\text{in } a') \quad u_1(c - hc_{-1}, l) &= \beta V_1(a', c) - \beta V_2(a', c) \\ (\text{in } l) \quad wu_1(c - hc_{-1}, l) + u_2(c - hc_{-1}, l) &= -w\beta V_2(a', c) \end{aligned} \quad (1)$$

- We derive the envelope conditions using the state variables a, c_{-1} ¹ :

$$\begin{aligned} (\text{in } a) \quad V_1(a, c_{-1}) &= (1 + r)u_1(c - hc_{-1}, l) + \beta(1 + r)V_2(a', c) \\ (\text{in } c_{-1}) \quad V_2(a, c_{-1}) &= -hu_1(c - hc_{-1}, l) \end{aligned} \quad (2)$$

- We iterate (2) over a, c_{-1}, r, l :

$$\begin{aligned} V_1(a', c) &= (1 + r')u_1(c' - hc, l') + \beta(1 + r')V_2(a'', c') \\ V_2(a', c) &= -hu_1(c' - hc, l') \end{aligned} \quad (3)$$

- We combine (1) and (3) and get the following expression:

$$\begin{aligned} u_1(c - hc_{-1}, l) &= \beta[(1 + r')u_1(c' - hc, l') + \beta(1 + r')V_2(a'', c')] - \beta V_2(a', c) \\ &= \beta[(1 + r')u_1(c' - hc, l') + \beta(1 + r')hu_1(c'' - hc', l'')] + \beta hu_1(c' - hc, l') \end{aligned} \quad (4)$$

- Hence, the EE is as follows:

$$EE : u_1(c - hc_{-1}, l) - \beta hu_1(c' - hc, l') = \beta(1 + r')[u_1(c' - hc, l') - \beta hu_1(c'' - hc', l'')]$$

- similarly, we get the following equation for the work-leisure tradeoff :

$$W/L : u_2(c - hc_{-1}, l) = w[\beta hu_1(c' - hc, l') - u_1(c - hc_{-1}, l)]$$

Hence, the following pricing kernel :

$$m' = \beta \frac{u_1(c' - hc, l') - \beta hu_1(c'' - hc', l'')}{u_1(c - hc_{-1}, l) - \beta hu_1(c' - hc, l')} = \frac{1}{1 + r'}$$

¹ l_{-1} could also be considered as a state variable however it does not enter directly the policy function so we omit it

3.3 3)

At the steady state, there are no variations across time so based on the EE we get :

$$u_1(c - hc, l)(1 - \beta h) = \beta(1 + r)u_1(c - hc, l)(1 - \beta h) \Leftrightarrow 1 = \beta(1 + r)$$

$$\Leftrightarrow r = \frac{1}{\beta} - 1$$

3.4 4)

We wish to find the steady state optimal labor supply assuming that $a_t = 0$. To this end we can use the work-leisure tradeoff equation we obtained in (2), let's derive $u_1(c_t - hc_{t-1}, l_t)$ and $u_2(c_t - hc_{t-1}, l_t)$:

$$u_1(c_t - hc_{t-1}, l_t) = (c_t - hc_{t-1})^{-\sigma}$$

$$u_2(c_t - hc_{t-1}, l_t) = -l_t^\epsilon$$

So we get the following equation for the work-leisure tradeoff :

$$-l_t^\epsilon = w[\beta h(c_{t+1} - hc_t)^{-\sigma} - (c_t - hc_{t-1})^{-\sigma}]$$

Now we take the steady state so all time indexes disappear and we get :

$$-l^\epsilon = w(\beta h - 1)(c - hc)^{-\sigma}$$

$$\Leftrightarrow l^\epsilon = w(1 - \beta h)(c - hc)^{-\sigma}$$

$$\Leftrightarrow l^\epsilon = w(1 - \beta h)c^{-\sigma}(1 - h)^{-\sigma}$$

Since $c = wl$ at the steady state based on the budget constraint with $a = 0$, we get :

$$l^\epsilon = w(1 - \beta h)w^{-\sigma}l^{-\sigma}(1 - h)^{-\sigma}$$

$$\Leftrightarrow l^{\epsilon+\sigma} = (1 - \beta h)w^{1-\sigma}(1 - h)^{-\sigma}$$

the optimal labor denoted l^* is thus as follows :

$$l^* = \left[\frac{(1 - \beta h)}{(1 - h)^\sigma} w^{1-\sigma} \right]^{\frac{1}{\epsilon+\sigma}}$$

Let's derive this expression in h to get the variation of l^* as a function of h :

$$\frac{dl^*}{dh} = \frac{l^*}{\epsilon + \sigma} \left(\frac{\sigma}{1 - h} - \frac{\beta}{1 - \beta h} \right)$$

The response to a change in habit h is ambiguous as it depends on σ and other parameters.

4

Exercise III

Please find the associated notebook at the following link : Notebook for Exercise 3

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Reference

Vasco M Carvalho, Makoto Nirei, Yukiko U Saito, Alireza Tahbaz-Salehi, Supply Chain Disruptions: Evidence from the Great East Japan Earthquake, The Quarterly Journal of Economics, Volume 136, Issue 2, May 2021, Pages 1255–1321